Date: Oct 07, 2026
To: "Sabri Manai" sabri.manai@doctoral.uj.edu.pl
From: "Future Generation Computer Systems" support@elsevier.com
Subject: Decision on submission to Future Generation Computer Systems

Manuscript Number: **FGCS-D-26-04298**

Conversational Counterfactual Guidance for Operator-Centred Hyperparameter Optimisation in Industrial Anomaly Detection

Dear Mr Manai,

Thank you for submitting your manuscript to Future Generation Computer Systems.

I have completed my evaluation of your manuscript. The reviewers recommend reconsideration of your manuscript following major revision. I invite you to resubmit your manuscript. Please resubmit your revised manuscript by **Oct 28, 2026**.

To view your reviewer feedback, please log in as an author at https://www.editorialmanager.com/fgcs/ and navigate to your manuscript in the " Submissions Needing Revision " folder under the Author Main Menu.

When revising your manuscript, please consider all issues mentioned in the reviewers' comments carefully. Please note that your revised submission may need to be re-reviewed.

Research Elements (optional)
This journal encourages you to share research objects - including your raw data, methods, protocols, software, hardware and more – which support your original research article in a Research Elements journal. Research Elements are open access, multidisciplinary, peer-reviewed journals which make the objects associated with your research more discoverable, trustworthy and promote replicability and reproducibility. As open access journals, there may be an Article Publishing Charge if your paper is accepted for publication. Find out more about the Research Elements journals at https://www.elsevier.com/authors/tools-and-resources/research-elements-journals?dgcid=ec_em_research_elements_email.

Future Generation Computer Systems values your contribution and I look forward to receiving your revised manuscript.

Kind regards,
Fred Suter
Special Content editor

Future Generation Computer Systems

Editor and Reviewer comments:

Reviewer #1: This manuscript presents CF-XHPO, a framework that reuses hyperparameter optimisation logs to provide operator-centred counterfactual configuration guidance for industrial visual anomaly detection, combining surrogate modelling, constrained retrieval, counterfactual generation, and a schema-constrained conversational interface, evaluated on the MVTec AD benchmark and the Proditec production dataset with PatchCore as the main model. The topic is timely and relevant to the special issue on data-driven innovation, and the paper addresses a genuine gap in the HPO literature by shifting focus from finding the best configuration to supporting post-hoc what-if reasoning under operator-defined practical constraints. The separation of surrogate-level query satisfaction from main-model confirmation is a methodologically sound design choice, and the honest reporting of the MVTec failure alongside the Proditec success strengthens the paper's credibility. The score-level diagnosis in Section 4.7, which localises the confirmation gap to the validation side and identifies pooled category score-scale incompatibility as a deployment bottleneck, is a genuinely informative analysis that goes beyond typical aggregate reporting. The 22-participant user study and the 40-utterance labelled parser evaluation add practical grounding. However, several substantial issues must be addressed before the manuscript can be considered for publication, warranting a major revision. First, the related work section, while covering four relevant threads, remains insufficiently connected to the broader machine learning modelling literature that underpins surrogate-based approaches; specifically, the authors should Enhance connections to recent literature that demonstrates the great potential and use of several machine learning models, such as the Gaussian process regression (doi: 10.1177/03019233241297720), graphical technique (doi: 10.1007/s11135-026-02740-y), neural network (doi: 10.1142/S2737599425500367), and ensemble/composite

method (doi: 10.1016/j.mlwa.2021.100035), for modelling complicated (nonlinear) patterns across a broad variety of study subjects in order to further motivate the exploration of machine learning models in your present work. The current justification for selecting CatBoost over the thirteen benchmarked regressors rests primarily on native categorical handling and built-in interpretability tooling, but a deeper discussion of why gradient-boosted trees were preferred over, for instance, Gaussian process surrogates—which offer calibrated uncertainty estimates directly relevant to the selection-conditioned fidelity problem the authors themselves identify—would substantially strengthen the methodological rationale. Second, the evaluation metrics, while including RMSE, MAE, and R^2 for surrogate fidelity and F1/AUROC for model confirmation, could be broadened; the authors should Use or at least discuss other performance measures, such as the relative root mean square error, that facilitate performance comparisons across models or targets (see doi: 10.1007/s43674-024-00075-5), particularly given that the two datasets exhibit different F1 ranges and the paper explicitly compares surrogate behaviour across MVTec and Proditec, making scale-normalised error metrics especially pertinent for cross-dataset interpretability. Third, the MVTec evaluation is restricted to four of fifteen categories, and while the authors justify this as a compute-budget decision, the resulting pooled score-scale pathology is inherently tied to this multi-category pooling; the paper would benefit from at least a brief single-category confirmation experiment to disentangle whether the surrogate error of −0.204 persists when the pooling problem is removed, since the upper-quantile experiment in Section 4.9 hints at this but does not isolate it cleanly. Fourth, the user study, while valuable, has notable limitations that the authors acknowledge but that nonetheless constrain the strength of the claims: 22 computer-science master's students are a proxy population, the sample imbalance between conditions (10 versus 12) limits any inferential statistics, the study-time parser achieved only 5% exact full-query accuracy so the results reflect the complete repair-and-iteration workflow rather than the conversational interface per se, and no demographic data were collected; the authors should discuss more explicitly how these limitations affect the generalisability of the reported ease-of-comparison finding and whether the observed difference in questionnaire scores (4.36 versus 3.44 for option comparison) could be attributable to the tabular-plus-textual presentation format rather than to the conversational interaction itself. Fifth, the conversational parser evaluation reveals that relative constraints fail entirely at both front-ends (0% exact accuracy on the relative subset), yet the framework's motivating use case—expressing goals such as "10% less review effort at similar quality"—depends precisely on such relative specifications; the paper should more clearly delineate what fraction of realistic operator queries would currently trigger the fallback and discuss the practical implications for deployment readiness, rather than treating this solely as a future-work item. Sixth, the retrieve-first policy recommendation in Section 4.9 is supported by the tested dense logs but the authors do not quantify what "adequate coverage" means operationally; a brief analysis of how retrieval performance degrades as log density decreases (for example, by subsampling the log to 50% or 25%) would transform the coverage-conditioned caveat into actionable guidance. Seventh, the paper's claim of sustainability and efficiency should be tempered: the complete-interaction cost of 141-253 seconds for eight candidates is longer than a single PatchCore retraining, and the amortised screening advantage of 1.02-1.74× on MVTec is marginal; the authors should discuss under what query frequencies or candidate counts the framework becomes clearly advantageous over sequential retraining, and whether the ACFX generation time of approximately 100-111 seconds per candidate is acceptable in an interactive operator workflow. Eighth, the explanation faithfulness issue acknowledged in Section 6.3 is non-trivial: since the conversational interface presents textual trade-off summaries that have not been audited against the structured payload, there is a risk of misleading operators; the authors should at minimum provide a qualitative example showing agreement or disagreement between the generated prose and the underlying data, or explicitly flag all explanations as unvalidated in the interface output. Ninth, the writing is generally clear but dense; several sections (particularly 4.7 and 4.9) pack multiple distinct findings into single paragraphs, and the paper would benefit from clearer sub-structuring or summary sentences at the end of each diagnostic subsection to aid readers navigating the technical evaluation. Tenth, the paper should clarify the relationship between the review-budget thresholding rule and standard operating-point selection in anomaly detection, as the former is presented as novel but closely mirrors existing top-k inspection protocols in industrial quality control; positioning this contribution more precisely relative to that literature would avoid overclaiming. In summary, the manuscript presents a well-motivated framework with a rigorous confirmation protocol and honest negative-result reporting, but requires strengthened literature positioning, broader evaluation metrics, additional diagnostic experiments to isolate the pooling effect, more explicit discussion of deployment readiness given the parser limitations, and clearer operational guidance for the retrieve-first policy before it meets the standard for publication in Future Generation Computer Systems.

Reviewer #2: Overall, I find the paper useful (and it studies an interesting problem) and the idea is interesting-the authors report both positive and negative results, the writing is generally clear and the numerical reporting seems consistent. There are still issues with the interpretation of the main MVTec result and with the surrogate analysis (particularly with the user study), the authors may be able to address these. For now, I would say Major Revision.

Major comments

1, the primary explanation for the MVTec failure is still an open question.

Fidelity conditioned on selection is the result mentioned in the paper, but there might be other reasons. For instance it is possible that the best HPO trials already have too optimistic validation accuracies or that the reported scores cannot be reproduced after retraining. In the case of LogTopK trials it would be important to check which recorded F1 score is truly reached upon retraining, the analysis should compare the recorded score with the rerun score across trials. All claims in the abstract and conclusions should be stated more carefully until this point is tested.

2, the high surrogate R^2 indeed requires more explanation.

While $R^2 = 0.995$ seems to be high, a large part of F1 can be directly explained by review budget, it is still unknown how much information can be obtained from hard hyperparameters. In this respect, I would appreciate if you could report feature importance or SHAP value, in addition to R^2 without budget, the other interesting question is the R^2 within each of the review budget bands.

3, the implementation of PatchCore needs to be tested.

The reported AUROC for MVTec is significantly below the published PatchCore results-in this paper their own implementation is used-I request the authors to add a run of their code/scoring with no HPO (i.e., simply with the paper's default 'reference configuration') on the chosen MVTec categories. That way one can determine whether the negative result on MVTec arises from the recommendation method itself or from the current implementation.

4. Some of the comparisons currently rely on a single run per candidate, which seems insufficient.

Very often the reported method differences are rather small, at the same time, the reported rerun delta may be of order 0.019. It would be helpful to repeat at least some subset of candidates with different seeds to check the variability, they should also avoid ranking methods too strongly when the differences are within that noise.

5, it is not a controlled comparison, e.g. MVTec (and Proditec).

The comparison should not be presented as (mainly) a benchmark-vs.-production comparison, because different objectives are used in both datasets and there are multiple differences including different objectives, different sizes of the log, different anchor and different structures (including the fact that MVTec pools a wide range of categories and Proditec is for one product line). A single category MVTec control or use of per-category objective would make the conclusion clearer.

6, the performance on MVTec differs according to the anchor.

It looks odd that one factual anchor accounts for almost all of the main negative finding, while another one in an upper quantile region apparently behaves much better, the number of confirmation candidates should be increased and include those collected with at least two more anchors. In addition, report how the verification rate changes and how the original anchor was picked.

7, the claim about the conversation is stronger than what the experiment shows.

During the user study the parser had low exact accuracy and used fallback often, the study therefore evaluated the full workflow including repair and fallback, not only the natural language parser, this should be clarified in the abstract and conclusion. Finally, over-reliance was not measured, although this is essential for a trustworthy conversational system.

8. More details are needed for the candidate-generation step and one simple baseline is absent.

Additional information is required on ACFX and the way DiCE is used. Please explain the way in which the continuous nature of F1 is treated and which implementation of ACFX has been used. Moreover, one should compare with a similar baseline that makes it possible to filter the existing HPO log for the query and which ranks candidates by their recorded F1, this is important because, ultimately, the paper suggests that a retrieve first approach is used.

Minor comments

Eq. 3, how do you deal with ties when multiple samples have the same anomaly score?

The "corruption level" is applied by the experimenter to the training images; I might rename this so as to avoid potential confusion with an operator action.

More information on the dataset from Proditec would be useful (such as image sizes, defect classes, and acquisition settings).

For Table 4, is this time per candidate or for all eight candidates? Please clarify, this needs to be clarified.

In Table 4, please note that ± should represent across candidates variation, not repeated run variation.

I would like the surrogate-versus-confirmed plot from MVTec as this is the more important case for the paper.

Please verify the odd value suggested by Table 10 for Condition A, Task 1, possibly it is a logging issue.

Figure 3 is drawn with a truncated y-axis and shows no spread. Could you present single points or error bars?

Section 4.6 performs a regression using 24 generated candidates that are not independent, so be careful about inferring a p-value.

Since the pooled distance correlation is predominantly a between-method effect, the pooled p-value is not appropriate

as the main result.

Please offer the simple "filter and sort" baseline in the runtime/cost comparison.

In case the article deals with LLM-as-a-judge, please add model, version, temperature, and prompt used in the Methods section.

Please spell out the ethical basis on which the student study was conducted with reference to pressure from staff.

Recent works relevant to related work are HPO-explanation and LLM-for-AutoML/HPO studies.

I would suggest the main issues of the research be clearly indicated at the end of the Introduction.

More information and support

FAQ: How do I revise my submission in Editorial Manager?
https://service.elsevier.com/app/answers/detail/a_id/28463/supporthub/publishing/

FAQ: How can I reset a forgotten password?
https://service.elsevier.com/app/answers/detail/a_id/28452/supporthub/publishing/kw/editorial+manager/

For further assistance, please visit our customer service site:
https://service.elsevier.com/app/home/supporthub/publishing/. Here you can search for solutions on a range of topics, find answers to frequently asked questions, and learn more about Editorial Manager via interactive tutorials. You can also talk 24/7 to our customer support team by phone and 24/7 by live chat and email.

At Elsevier, we want to help all our authors to stay safe when publishing. Please be aware of fraudulent messages requesting money in return for the publication of your paper. If you are publishing open access with Elsevier, bear in mind that we will never request payment before the paper has been accepted. We have prepared some guidelines (https://www.elsevier.com/connect/authors-update/seven-top-tips-on-stopping-apc-scams) that you may find helpful, including a short video on Identifying fake acceptance letters (https://www.youtube.com/watch?v=o5l8thD9XtE). Please remember that you can contact Elsevier s Researcher Support team (https://service.elsevier.com/app/home/supporthub/publishing/) at any time if you have questions about your manuscript, and you can log into Editorial Manager to check the status of your manuscript (https://service.elsevier.com/app/answers/detail/a_id/29155/c/10530/supporthub/publishing/kw/status/).

#AU_FGCS#

To ensure this email reaches the intended recipient, please do not delete the above code

5

In compliance with data protection regulations, you may request that we remove your personal registration details at any time. (Remove my information/details). Please contact the publication office if you have any questions.